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A Scoping Survey of Quantum Machine Learning and Deep Learning for Real-World Applications

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Abstract

Many prevalent issues in today's society, such as fake news detection, can be efficiently addressed using artificial intelligence and machine learning techniques. The rapid dissemination of fake news through social media makes it challenging to verify the validity of information. QML and QDL represent promising frontiers for building models to address these challenges, with the prospect of achieving quantum advantage driving advancements in the field. This study examines and compares literature on QML and QDL models, including QSVM, VQE, QNN, QCNN, and RQNN, using the MNIST dataset. The effectiveness and scope of application of these models for fake news detection and other real-world applications are analyzed.

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1. Introduction

Over the last decade, AI and ML have evolved into key transformative technologies, witnessing rapid advancements across both theoretical and practical domains. These fields enable the processing of large-scale data and the generation of predictive insights, rendering them indispensable in numerous real-world systems, such as healthcare diagnostics, autonomous systems, and financial modeling [1, 2]. Classical AI approaches have substantially optimized tasks that were previously manual or inefficient; however, significant computational bottlenecks are arising, particularly in scalability, processing speed, and resource consumption [3].

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The advent of Quantum Computing (QC) introduces the potential to overcome these limitations, providing the foundation for Quantum Machine Learning (QML) and Quantum Deep Learning (QDL) [4]. Unlike its classical counterpart, quantum computing leverages the principles of quantum mechanics to perform computations, enabling parallel processing of large datasets through quantum superposition and entanglement. The era of quantum advantage, where quantum computers surpass classical computers in solving specific problems, is approaching, and By merging Quantum Computing and Machine Learning, AI systems could see substantial improvements in efficiency and processing speed, leading to groundbreaking applications [5].

A core advantage of QML lies in its ability to significantly reduce the training time for complex models. Classical AI systems, though effective, require substantial time and computational resources to train deep neural networks and other models, particularly when handling high-dimensional data. In contrast, quantum computers have demonstrated the potential to accelerate these processes by several orders of magnitude. It is estimated that QML algorithms can outperform classical ML algorithms by up to 100x, with even more significant speedups possible in certain cases [6]. This represents a key milestone towards achieving quantum supremacy—the point at which quantum machines outperform classical systems in all but trivial tasks [7].

Effectively harnessing the power of quantum systems requires representing classical data in a quantum-compatible format. Encoding classical data into quantum formats, such as amplitude encoding or phase encoding, allows for exponential data representation. This method enables more efficient operations on massive datasets using fewer quantum bits (qubits) than classical bits [8]. Quantum systems' ability to encode and manipulate large amounts of information with fewer resources offers distinct advantages over classical methods, especially for tasks involving high-dimensional and complex data structures.

A fundamental component of QML is the use of Parameterized Quantum Circuits (PQC). PQCs consist of quantum gates with tunable parameters optimized during training, similar to how weights in classical neural networks are optimized. The entanglement between qubits in PQCs facilitates the exploration of complex data relationships, enabling the discovery of patterns that would be computationally prohibitive for classical systems [9]. Optimization of these parameters through quantum gradient descent or other quantum-enhanced methods leads to faster convergence and more accurate models.

This paper offers a comprehensive analysis of QML and QDL models, focusing on the computational advantages in terms of speed, model complexity, and data efficiency. Applications in real-world scenarios, such as financial modeling, cryptography, and drug discovery, are also explored [10]. A comparative analysis will evaluate the performance of quantum models against classical ML approaches, highlighting the benefits that quantum computing brings to machine learning tasks [11].

2. Related Works and Basic Concepts

In recent years, significant research has been conducted in the domain of quantum computing and its applications in machine learning [12]. The idea of using quantum mechanics to enhance machine learning algorithms is not entirely new but has gained tremendous attention as quantum hardware advances. Quantum Machine Learning (QML) represents an amalgamation of quantum computation and traditional machine learning, promising superior computational speeds and efficiencies compared to classical counterparts [5].

Several studies have laid the foundation for QML by exploring different quantum algorithms that can be applied to machine learning problems [13]. One of the pioneering approaches was the Quantum Support Vector Machine (QSVM) [11], which leverages quantum computing to speed up classification tasks, as demonstrated in the works of [14]. Similarly, quantum variants of clustering algorithms have been proposed, showing faster processing times compared to classical clustering techniques [15]. Additionally, the potential of quantum neural networks (QNNs) has been widely discussed, particularly for tasks involving deep learning, where quantum principles can optimize training times and model convergence [16].

Another critical area in QML research is the encoding of classical data into a quantum-compatible format. Methods such as amplitude encoding and phase encoding [13] have been employed to represent classical data with fewer quantum bits (qubits), allowing for more efficient computations [8]. These encoding techniques have been integral in improving the efficiency of quantum machine learning algorithms. The development of Parameterized Quantum

Circuits (PQC) [11], which are essential for many QML models, has been a topic of significant exploration as well. PQCs utilize entangled qubits to identify complex patterns in data, leading to improved learning models [9].

In the context of real-world applications, quantum algorithms have shown promise in fields like cryptography, drug discovery, and optimization tasks. For instance, quantum-enhanced reinforcement learning has been applied to optimize decision-making systems in financial markets [17], while quantum algorithms have been used to simulate molecular interactions in drug discovery, significantly reducing the time required for these simulations [18].

2.1. Basic Concepts of Quantum Machine Learning

As quantum computing and learning differ from classical computing in some of the key concepts, it is important to establish a basic understanding of the underlying concepts, namely the qubit, product states, entanglement, and parameterized circuits, before attempting to draw comparisons in the models.

2.1.1. Qubit

The qubit is the unit of storage in quantum computers that records data, but it differs from the classical bit, which can only be in discrete states, either 0 or 1 [13]. A qubit, in theory, can occupy a superposition of states, meaning it can be in both a 0 and 1 state simultaneously, and upon measurement, can produce either value with a probability [19]. The value of a qubit is represented using a column vector of probability amplitudes, which represent the probability of each state being measured. The amplitudes can be negative, and the sum of squares of all amplitudes is always 1. A Hadamard gate can be used to place a qubit into an equal superposition of 0 and 1. Other unitary gates modify the probability amplitudes of qubits, and through different superpositions, it becomes possible to represent more data.

2.1.2. Data Encoding

Unlike classical bits, qubits can represent more data. Encoding classical data can be done using simple binary strings, or the data can be translated to accommodate the probability amplitudes of the qubit. Data may even be encoded in the angles of amplitude in the qubit [20, 19]. Quantum data can also be used directly.

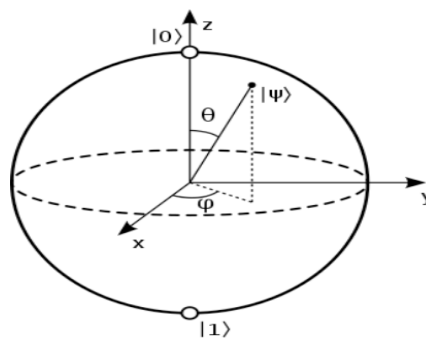


Fig. 1: Bloch Sphere of qubit

2.1.3. Product State and Entanglement

When considering two or more qubits in a circuit, their combined possible states can also be represented in a column vector. For two qubits, there are four possible states, each with its probability of being measured. A superposition of states where the qubits' states directly contribute to the combined state, without dependence on other qubits, are known as product states, behaving similarly to classical bits. When the combined superposition cannot be represented as a combination of individual states, the qubits are considered entangled. Entanglement has various applications in quantum computing, such as in the teleportation protocol in communication and quantum learning. Entanglement is often achieved using two-qubit gates like the controlled NOT (CX) gate. Gates such as the Hadamard or CX gate have defined state transitions, represented by square matrices called Pauli matrices, which describe operations on qubits through state transitions.

2.1.4. Parameterized Circuit

In quantum machine learning, models are often represented as circuits with gates containing tunable parameters (θ) [19]. These parameters can be trained and optimized using the gradient descent approach, which involves calculating a cost function and adjusting the parameters by a multiple of its derivative to the parameter and the learning rate, a defined hyperparameter. This is analogous to classical machine learning. In quantum machine learning, the cost function typically involves the expectation value, which is the average of the expected state values of the parameter vector.

3. PARAMETERS FOR COMPARISON

Having established the core concepts behind QML and QDL, the parameters for performing the analysis are decided by identifying the factors that affect the performance and viability of the model for real-world applications. This is done by focusing on three key questions:

1. What primarily affects the size of the model?
2. What affects the speed of the model?
3. What affects the accuracy of the model?

Upon reviewing the models and research on QML and QDL, the following conclusions are drawn: The size of the model is primarily affected by the size and complexity of the dataset, as well as the number of qubits available for the model. This allows for ideal fitting, as complexity requires more layers of optimizer PQCs before obtaining results, and the number of qubits limits the scope. Therefore, the goal for a QML or QDL model is to achieve a balance in the number of optimizer PQCs required over the number of qubits available. An effective model will use a moderate number of gates on fewer qubits, allowing for scalability. The speed of the model is primarily influenced by the nature of qubits and the required circuit for the model, and is more dependent on architecture than development design. For comparison purposes, it is assumed that the architecture of the models is identical and does not affect speed, so the number of epochs required becomes relevant. Finally, the accuracy of the model depends on the complexity and degree of entanglement, or the number of gates in the optimizer PQC. This leads to increased size, so it is important to maintain feasibility between size, speed, and accuracy. The number of qubits, gates (if mentioned), epochs, and the achieved accuracy and F1 score presented in the research will serve as the comparison parameters moving forward.

4. Models for Comparison

This survey considers research on QML models like QSVM and VQT and QDL models such as QNN, QCNN, and RQNN. The goal is to provide a precise summary of each model and its characteristics and compare them to one another to determine their advantages.

4.1. QSVM

4.1.1. Model Explanation

QSVM and its variant QSVM-Kernel utilize quantum circuits to efficiently handle machine learning tasks. By mapping classical data points nonlinearly to a high-dimensional feature space using N qubits [13], QSVM enables classification based on separation hyperplanes derived from the training phase [21]. Comparative evaluations against classical SVM classifiers have demonstrated QSVM's process, particularly evident in applications like handwritten digit recognition on the MNIST dataset. QSVM exhibits significant speed advantages and improved accuracy, making it a promising candidate for accelerating complex machine learning problems.

4.1.2. Experimentation

A QSVM architecture configured with 14 qubits and 32 parameters was employed. The training process spanned 50 epochs, and the accuracy was meticulously recorded after every 20 batches [22]. Table 1 below presents the mean and standard deviation of the classification accuracy obtained using this QSVM setup with the MNIST dataset.

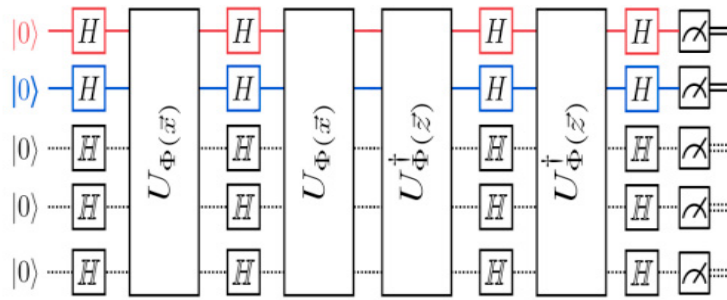


Fig. 2: Structure of QSVN Network

Table 1: Experimental results - QSVN

Algorithms	QSVN
Qubits	14
Parameters	32
Readouts	1
Best accuracy (%)	$84.68 \pm 0.6\%$
Best F1 score	0.8878

4.1.3. Applications of QSVN

Beyond digit recognition, QSVN finds applications in image recognition, pattern detection, and quantum-classical hybrid models. Its ability to handle high-dimensional feature spaces and leverage quantum computing capabilities opens avenues for tackling real-world problems across domains [23].

4.2. VQT

4.2.1. Model Explanation

VQT offers a novel approach to simulating complex thermal states using variational quantum circuits [24]. Leveraging the variational principle, VQT optimizes quantum circuits iteratively to approximate thermal equilibrium states of a given Hamiltonian. Demonstrating efficiency in modeling thermal systems, VQT outperforms classical methods in certain regimes. Its applications span condensed matter physics, chemistry, and material science, providing insights into phase transitions, energy distributions, and correlation functions within quantum systems.

4.2.2. Experimentation

A VQT architecture configured with 12 qubits and 16 parameters was employed. The training process spanned 50 epochs, and the accuracy was meticulously recorded after every 20 batches. Table 5 below presents the mean and standard deviation of the classification accuracy obtained using this VQT setup with the MNIST dataset.

Table 2: Experimental results - VQT

Algorithms	VQT
Qubits	12
Parameters	24
Readouts	1
Best accuracy (%)	$86.16 \pm 0.4\%$
Best F1 score	0.8228

4.2.3. Applications of VQT

VQT's potential applications extend to material design, fundamental physics research, and the exploration of emergent phenomena in quantum systems [25]. As quantum computing technology advances, VQT stands poised to become a valuable tool for understanding complex thermal behavior and designing novel materials.

4.3. QNN

4.3.1. Model Explanation

Quantum Neural Networks (QNNs) merge quantum computing with neural networks, offering new computational possibilities. Unlike classical neural networks that use bits (0 or 1), QNNs use qubits, which can exist in multiple states simultaneously due to superposition. QNNs utilize quantum gates to process information, exploiting quantum properties like superposition and entanglement [11] to perform computations that might be more efficient or parallel compared to classical methods. While QNNs aim to learn and predict patterns, much like traditional neural networks, their full potential depends on advances in scalable quantum computing technology.

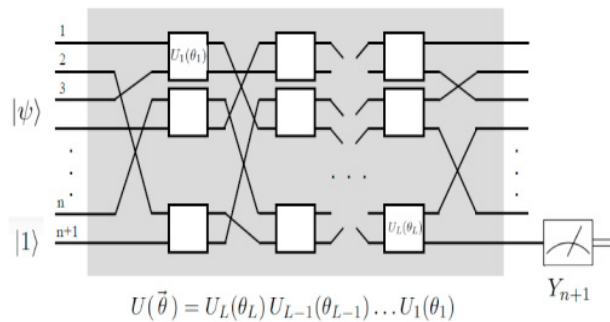


Fig. 3: Structure of QNN circuit [26]

4.3.2. Experimentation

The model was set up with 8 Qubits and 16 parameters. It was trained over 20 epochs, and the accuracy was recorded for every 32 batches. In this work, the mean and the standard deviation of the classification accuracy using the QNN architecture with the MNIST dataset are given in the following table [27, 28].

Table 3: Experimental results - QNN

Algorithms	QNN
Qubits	8
Parameters	16
Readouts	1
Best accuracy (%)	82.88 ~ 1%
Best F1 score	0.8378

4.3.3. Applications of QNN

Reinforcement Learning: Deep learning has increasingly been combined with reinforcement learning to optimize actions in complex state spaces. Integrating quantum deep learning into reinforcement learning is still in its infancy. The current approach replaces the policy training network with a Quantum Neural Network (QNN) [29] instead of a traditional deep neural network. Classical deep reinforcement learning algorithms may be adapted for quantum systems. If QNNs demonstrate quantum computational advantages in complex environments like intricate Markov decision processes, quantum reinforcement learning could significantly advance the field.

Communication Networks: Quantum Neural Networks (QNNs) and quantum reinforcement learning techniques are widely applicable across diverse domains, such as communication networks [29]. In decentralized systems like blockchain, QNNs can potentially speed up computational processes significantly. Furthermore, modern communication technologies, including the Internet of Things (IoT), millimeter-wave networks, caching infrastructures, and video streaming [29] platforms, present valuable opportunities for implementing QNNs and quantum reinforcement learning algorithms.

4.4. QCNN

4.4.1. Model Explanation

Quantum computing uses quantum mechanics principles for computation, and Quantum Convolutional Neural Networks (QCNNs) integrate these concepts into neural networks. Unlike classical bits, qubits in quantum computers can exist in multiple states simultaneously, offering potential advantages for certain tasks. QCNNs apply quantum convolutional layers, using quantum gates to perform operations similar to classical CNNs, with the hope that quantum properties like superposition and entanglement will enhance optimization and pattern recognition[13].

While classical machine learning handles tasks like image recognition effectively, translating quantum problems into classical frameworks is challenging due to the exponentially large Hilbert space involved. Quantum algorithms can address these challenges, but the current limitations in quantum hardware [30], such as size and coherence times, restrict the deployment of large-scale quantum networks.

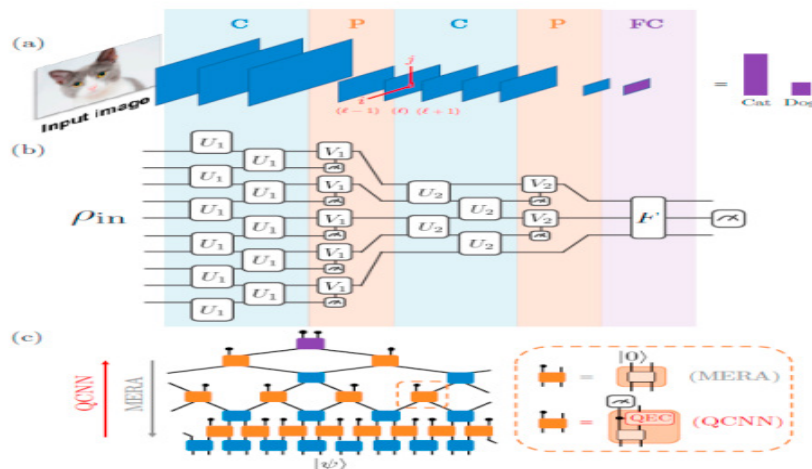


Fig. 4: Representation of CNN [28]

4.4.2. Experimentation

The model was set up with 8 Qubits and 63 parameters. It was trained over 20 epochs, and the accuracy was recorded for every 32 batches. The mean and standard deviation of the classification accuracy using the QCNN architecture with the MNIST dataset are given in Table [27].

4.4.3. Applications of QCNN

Computer Vision Applications: CNN is extensively used in image recognition and object detection for classification purposes. QCNN can increase computational speeds with better performance metrics compared to classical computational methods.[31]

Quantum Chemistry: Modeling molecular structures and simulating quantum systems for drug discovery or materials science could benefit from QCNNs [11]. These networks might provide more accurate representations of quantum states, allowing for better predictions and optimizations in quantum chemistry simulations [31].

Table 4: Experimental results - QCNN

Algorithms	QCNN
Qubits	8
Parameters	63
Readouts	1
Best accuracy (%)	93.17 ~ 1%
Best F1 score	0.9352

Financial Modeling: QCNNs might be applied to quantum-enhanced financial modeling for portfolio optimization, risk assessment, and other quantitative finance tasks. Quantum computing’s potential for handling complex optimization problems could be particularly beneficial in this domain [31].

Complex System Simulations: QCNNs could contribute to simulating and understanding complex quantum systems, surpassing the capabilities of classical computers in certain scenarios. This has implications for materials science, physics, and other fields where simulating intricate quantum behaviors is crucial [31].

4.5. RQNN

4.5.1. Model Explanation

Recurrent Neural Networks (RNNs) play a fundamental role in modeling sequential data for tasks like machine translation and speech synthesis. These networks utilize hidden memory cells that factor in previous inputs, with variations like LSTM and GRU [32] adjusting input influence. However, optimizing RNNs for long sequences remains challenging due to issues like vanishing or exploding gradients, which hinder training performance.

In contrast, quantum computing is still emerging, but quantum machine learning models like variational quantum eigensolvers (VQEs) [32] have shown promise. Building on this, researchers are developing Recurrent Quantum Neural Networks (RQNNs) to handle sequential data. The proposed RQNN model uses a VQE to optimize each input layer or iteration, with the cell state stored in additional qubits for further iterations, altering inputs with each pass [32].

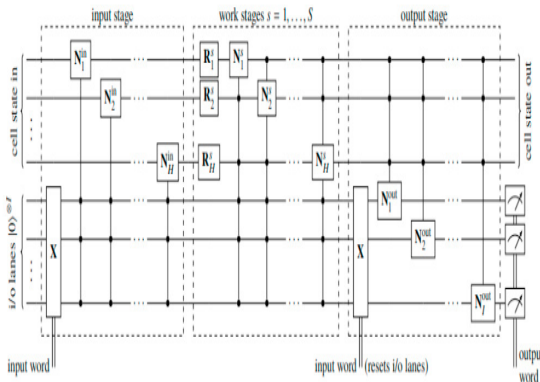


Fig. 5: Schematic representation of the proposed RQNN circuit [32]

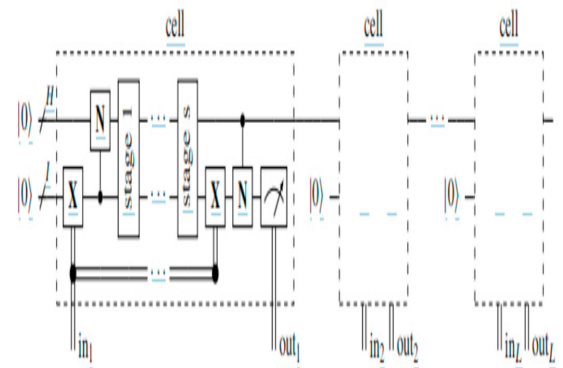


Fig. 6: QRNN, by applying the same QRNN cell constructed repeatedly to a sequence of input words. [32]

4.5.2. Experimentation

The QRNN was implemented in PyTorch, utilizing custom quantum gate layers to extract predicted distributions at each step [32]. Since this was a quantum computation simulation, certain shortcuts were applied: instead of using fixed-point amplitude amplification for quantum neurons and outputs during training, post-selection probabilities were manually selected and tracked to estimate the necessary overhead. Output probabilities were directly extracted, rather than being estimated through measurements at each step. The experiments focused on character-wise

RNNs, using various data preprocessing methods, including PCA. The predicted distributions were fed into PyTorch's nn—CrossEntropyLoss for gradient-based learning via PyTorch's auto-grad framework. On quantum hardware, alternative methods such as gradient-free optimizers or numerical gradients would be required. The experiments were executed on 2-8 CPUs, with memory usage ranging from 500MB to 35GB per core and parallel batch training. Running the QRNN on real hardware would involve parallel execution across multiple devices, averaging the losses [32].

Table 5: Experimental results - RQNN

Algorithms	RQNN
Qubits	10
Parameters	1292
Readouts	1
Best accuracy (%)	94.6 $\pm$ 0.4%
Best F1 score	-

4.5.3. Applications of RQNN

Recurrent Quantum Neural Networks find scope for application in any sequential data task and NLP application such as Sentiment analysis, Chatbots, Machine Translation and speech synthesis, as well as text analytics [33]. Real-time sentiment analysis tracks social media mentions, monitors feedback on marketing initiatives or product launches, and gives a general idea of how an audience feels about a business [32, 33]. Likewise, any application involving time-sequenced text, images, or media can be optimized and categorized using RNNs, with RQNNs speeding up large applications in these domains. Natural language processing has many use cases in the digital world, and this list will expand as more organizations and industries adopt it and recognize its benefits. While human interaction is crucial for complex communication challenges, NLP simplifies and automates simpler tasks before addressing more intricate ones [33].

5. QML SOFTWARE

This section reviews several key quantum computing resources, software, and tools for developing quantum machine learning algorithms.

- **Microsoft Azure Quantum:** Part of the Azure Cloud, Azure Quantum requires membership access. It allows users to develop quantum programs using Q# within the Quantum workspace and run them on simulators or partner quantum computers like IONQ, Quantinuum, and Rigetti. Azure Quantum integrates with other Azure services and supports Qiskit [19] programming. Q#, a language compatible with .NET, offers high-level abstraction and libraries for quantum algorithms, quantum chemistry, and machine learning [34]. Comprehensive resources, including documentation, tutorials, and a GitHub repository, support learning and development [35]. The classifier uses single-qubit rotations and two-qubit controlled rotations, with the learnable parameters being the rotation angles [36]. Users can design custom classifiers using quantum operations on circuit qubits [37].

- **IBM Quantum Computing Solutions for Enterprises:** IBM offers enterprise-level quantum computing solutions, adhering to IEEE standards for accuracy, reliability, and interoperability. The **IBM Quantum Optimization Solver** uses quantum annealing to enhance performance in solving complex optimization problems like supply chain logistics and portfolio management [11]. **IBM Quantum Machine Learning** integrates quantum capabilities into traditional machine learning workflows, enabling more accurate and efficient model training. **IBM Quantum Financial Services Solutions** provides specialized tools for portfolio optimization, risk management, and fraud detection in the financial sector, while **IBM Quantum Supply Chain Solutions** optimizes logistics and resource allocation using quantum methodologies.

- **IBM Quantum Computing Tools:** IBM provides tools like the **IBM Quantum Composer**, a graphical interface for designing quantum circuits, and **IBM Quantum Experience**, an online platform offering access to quantum processors, tutorials, and resources. **Qiskit**, IBM's open-source quantum computing framework, enables users to create, manipulate, and analyze quantum circuits, with extensive support for algorithm development and execution on quantum systems.

- **AWS Quantum Computing:** AWS Quantum Computing, through Amazon Braket, offers cloud-based access to quantum resources. It supports quantum-classical hybrid computing, and various quantum processors, and integrates with other AWS services. Despite current limitations in the Noisy Intermediate-Scale Quantum (NISQ) era, AWS explores applications like [38] **Quantum solutions for portfolio optimization**, leveraging algorithms like the Quantum Approximate Optimization Algorithm (QAOA) to potentially achieve computational speedup.

Algorithm: Quantum computers leverage unique properties of quantum mechanics, notably qubits in superposition, enabling simultaneous processing of numerous possibilities. Entanglement links qubit states regardless of distance, while interference manipulates quantum state amplitudes to alter outcome probabilities. Quantum algorithms, such as Shor's algorithm, demonstrate superior efficiency compared to classical counterparts, solving certain problems exponentially faster. The Quantum Approximate Optimization Algorithm (QAOA) blends classical and quantum computation, utilizing mixer and cost layers to explore potential solutions. Classically fine-tuned parameters iteratively converge during each QAOA run, moving towards optimal values. This hybrid approach capitalizes on quantum advantages while integrating classical optimization for enhanced performance.

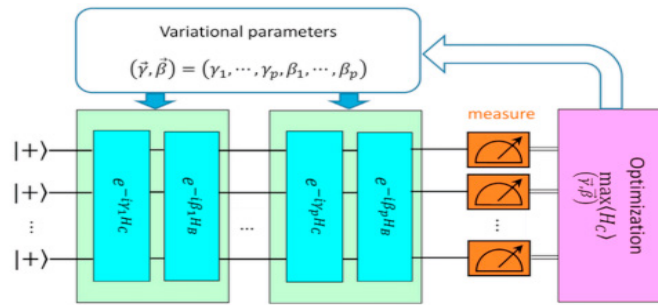


Fig. 7: A schematic representation of QAOA.

The circuit is composed of 'p' layers, each having a cost sub-layer function and a mixer sub-layer function. Variational parameters are changed with a classical optimizer after measuring the objective function following each iteration. The Quantum Approximate Optimization Algorithm (QAOA) is a versatile algorithm adept at solving optimization problems with numerous constraints. When incorporating constraints, two primary options exist. The first involves modifying the mixer layer to include only valid solutions, limiting the search space. The second option alters the cost layer by adding a penalty factor to penalize invalid solutions, producing shallower circuits suitable for near-term noisy quantum computers. This study focuses on implementing a penalty factor for portfolio optimization to increase the likelihood of selecting an optimal asset mix. Two constraint types, equality and inequality, are explored. For equality constraints, a defined budget μ is introduced, requiring the entire budget to be spent. A penalty factor coefficient σ is added to the portfolio optimization problem to ensure compliance.

$$\omega^{opt} = \arg \min_{\omega} \left(\frac{1}{2} \omega^T \Sigma \omega - \lambda R^T \omega + \sigma \left(\sum_i \omega_i - \mu \right)^2 \right) \quad (1)$$

For inequality constraints, an upper spending limit is set: $\sum \omega_i \leq \mu$. Introducing slack variables (s) eliminates penalty term factors for valid solutions, satisfying the inequality condition. Although adding slack variables increases qubit and operational step requirements, it results in shallower circuits compared to the alternative method of incorporating constraints.

$$\omega^{opt} = \arg \min_{\omega} \left(\frac{1}{2} \omega^T \Sigma \omega - \lambda R^T \omega + \sigma \left(\sum_i (\omega_i + s_i - \mu) \right)^2 \right) \quad (2)$$

This emphasis on shallow circuits is crucial, given the error susceptibility of current quantum computers, where errors accumulate with increasing circuit depth. By exploring penalty factors and constraints in portfolio optimization, this research aims to enhance the practicality and effectiveness of quantum algorithms for real-world financial applications on existing quantum computing hardware.

The analysis of the results [39] reveals that the results are sensitive to the penalty factor, which must be analyzed with different values to fit individual use cases.

- Optimization of Robot Trajectory Planning with Nature-Inspired and Hybrid Quantum Algorithms [20]:

Robot motion planning is critical in industries like automotive, manufacturing, and logistics, where optimizing robotic paths is essential for tasks such as welding, painting, and assembly. Efficient load balancing among numerous robots and task sequencing is crucial to meet production demands. Quantum computing, particularly quantum annealing devices from companies like D-Wave Systems Inc., offers promising solutions for these complex optimization problems, although the field is still evolving from research to practical applications. The main challenge is identifying the quantum hardware and algorithms that will provide a real-world advantage.

To address these challenges, optimization methods must bridge the gap until scalable quantum hardware becomes available, allowing for a seamless transition to quantum solutions when accessible. Current quantum hardware, especially annealers, utilizes the Quadratic Unconstrained Binary Optimization (QUBO) framework [13]. This framework models a wide range of NP-hard combinatorial problems, though it can require many variables for certain applications. The QUBO cost function is represented by the Hamiltonian:

$$H_{\text{QUBO}} = \mathbf{x}^T Q \mathbf{x} = \sum_{i,j} x_i Q_{ij} x_j \quad (3)$$

where $\mathbf{x}$ is a vector of binary decision variables, and Q is the matrix encoding the problem. To guide the solver towards feasible solutions, penalty terms are added to the Hamiltonian to enforce constraints, such as visiting exactly one node per time step and ensuring all seams are visited once during the tour.

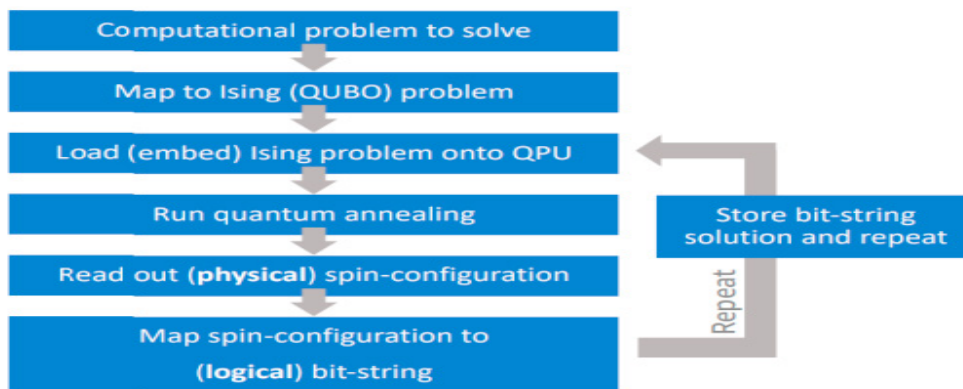
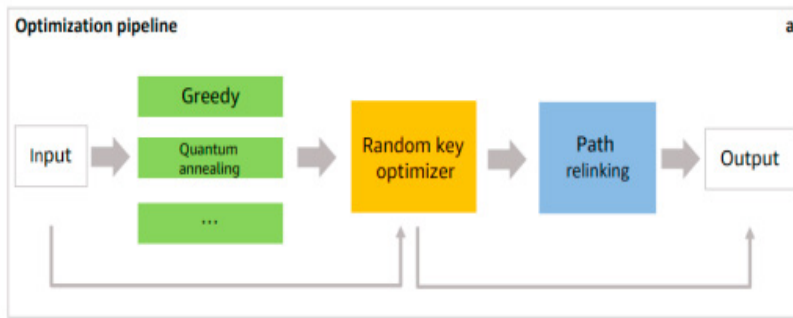


Fig. 8: Flow chart illustrating the end-to-end workflow for solving a combinatorial optimization problem on a quantum annealer.



To tackle optimization problems using quantum annealing, the problem is first formulated as a Quadratic Unconstrained Binary Optimization (QUBO) or Ising model [40]. This abstract problem is mapped onto the quantum processing unit (QPU), often requiring an increased number of variables due to the sparse connectivity of the quantum chip. Quantum annealing is then applied to identify an optimal or near-optimal solution, which is subsequently converted back to a bit-string representing the original problem. Due to the probabilistic nature of quantum annealing, the process is repeated multiple times, followed by statistical analysis to find the best configuration that minimizes the objective function.

Quantum computing leverages phenomena that extend beyond classical capabilities, with two primary paradigms: universal circuit-based quantum computers and specialized quantum annealers. Circuit-based devices offer the potential for exponential speed-ups but encounter scalability challenges, as error correction is required to achieve reliable logical qubits from the current limit of approximately 100 physical qubits. In contrast, quantum annealers are specifically designed for combinatorial optimization problems, particularly within the QUBO class, and currently utilize around 5000 analog superconducting qubits, providing a significant qubit advantage [41].

The optimization pipeline typically includes a core routine processing cost values between node pairs, directed into a random key optimizer (RKO) that employs heuristic search for an optimized solution. This core routine can be extended upstream with warm-start algorithms like quantum annealing to diversify the initial population and downstream with path-relinking techniques for refining high-quality solutions [42].

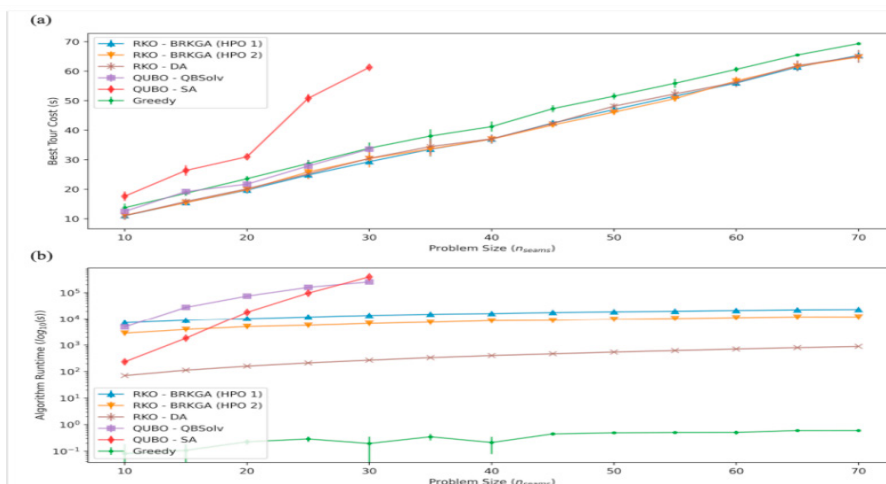


Fig. 9: Results analysis from [43]

Table 6: Comparison of Models

Model	QSVM	VQE	QNN	QCNN	RQNN
Qubits	14	12	8	8	10
Parameters	32	24	16	63	1292
Outputs	1	1	1	1	1
Accuracy (%)	84.68 $\pm$ 0.6%	86.16 $\pm$ 0.4%	82.88 $\sim$ 1%	93.17 $\sim$ 1%	94.6 $\pm$ 0.4%
F1 score	0.8878	0.8228	0.8378	0.9352	-
Scope of Application	Analysis of Relations in Data	Analysis of Relations in Data	Analysis of Relations in Data	Image Analysis, Tampering Detection	Natural Language Processing, False Text Detection

6. CONCLUSION AND FUTURE WORK

This paper has undertaken a detailed review aimed at analyzing QML and QDL algorithms, focusing on their applications and performance. The study of research implementations of quantum machine learning models has provided valuable insights into their scope, domain, and potential for real-world use cases. The analysis centered on evaluating the performance metrics of these models, particularly with the MNIST dataset, and identifying the strengths and weaknesses of each approach. Through comparative assessments, opportunities for further development and practical deployment of these models in various systems have been highlighted.

Future research will build upon these findings by refining quantum machine learning (QML) and quantum deep learning (QDL) models[4], with a specific focus on addressing the problem of fake news detection. This will involve enhancing datasets to include a wider range of authentic and deceptive news articles, rigorously annotated for classification. By iteratively optimizing QML/QDL models and fine-tuning parameters, the goal is to improve classification accuracy. Further, evaluations against traditional machine learning methods will offer deeper insights into the effectiveness of quantum-based approaches in combating misinformation. Exploring avenues for the real-world deployment of these optimized models could lead to their integration into practical applications, such as real-time fake news detection, underscoring the transformative potential of quantum computing in addressing critical societal challenges.

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